

Objective

Build a **multi-agent AI system** that can answer user queries using:

- Retrieval-Augmented Generation (RAG)
- Tool usage
- Basic agent orchestration

We recommend using **Agno Agent OS (UI optional)** and a free LLM via OpenRouter.

Problem Statement

Create an **AI Research Assistant** that can:

- Answer general questions
 - Retrieve and use information from a document knowledge base
 - Use tools when required (e.g., calculator, lookup)
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Requirements

1. Core Agent

- Uses an LLM (e.g., via OpenRouter)
 - Can:
 - Understand user intent
 - Decide when to use tools or RAG
 - Generate final responses
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2. RAG Pipeline

Implement a basic RAG system:

- Load documents (PDF, text, or markdown)
- Chunk and embed data
- Store DuckDB,
- Retrieve top relevant chunks

Important:

You must **show retrieved context** in your output (not just use it silently).

3. Tools (Minimum 2)

Examples:

- Calculator
- Document lookup
- Simple web search (can be mocked)

Agents should:

- Decide when to call tools
 - Use structured inputs/outputs
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4. Multi-Agent Setup

Implement at least:

- **2 agents + 1 coordinator**

Example:

- Retriever Agent (handles RAG)
 - General Agent (handles reasoning)
 - Coordinator (routes queries)
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5. (Optional but Recommended) UI

- Simple interface using Agno UI or CLI
 - Show:
 - Final answer
 - Which agent handled the query
 - Retrieved chunks (for RAG transparency)
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Deliverables

- GitHub repository
 - README including:
 - Architecture diagram
 - Design decisions
 - Tradeoffs
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Bonus (Optional)

- Query classification before routing
 - Avoid unnecessary RAG calls
 - Logging / tracing
 - Small evaluation dataset (5–10 queries)
 - Performance optimizations
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Evaluation Criteria

We will evaluate based on:

- **Agent Design** – clear reasoning vs prompt chaining
 - **RAG Quality** – relevant retrieval, not noise
 - **Tool Usage** – structured and meaningful
 - **System Design** – modular and clean
 - **Code Quality** – readability and organization
 - **Thinking** – depth of explanation in README
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Common Pitfalls

Avoid:

- Always calling RAG (no decision logic)
- Not showing retrieved context
- Hardcoded or shallow implementations
- Fake multi-agent setups (no real separation)

Note

Focus on **clarity of thinking and system design**, not just making something work.